

Lab session 2: Implementation of digital systems

– ADC, DAC, sampling principle and aliasing error

EMBEDDED SIGNAL PROCESSING SYSTEMS HT 2018

Instructions

- The lab sessions are intended for you to implement a real-world embedded DSP system based on Atmel UC3A3 Xplained evaluation kit, and learn to solve practical problems which you encounter. The embedded system is aimed at real-time digital signal processing (RT DSP) and consists of a computing unit, analog-to-digital converters (ADCs) and digital-to-analog converters (DACs), as well as USART for data communication between the MCU and a PC.
- This lab session is mainly focused on ADC, DAC, sampling principle and aliasing error, and also on the signal generation using MATLAB. The main tasks in this lab session are the following
 - (i). Observing how the output signal from the system changes (especially in frequency) when the input is a sine signal and its frequency is changed from 1 kHz – 50 kHz in two cases: (i) without and (ii) with an RC LP filter before the ADC
 - (ii). Understanding how the ADC and DAC work
 - (iii). Thinking about how you implement an embedded DSP system when you have evaluation kit
 - (iv). Generating signals using MATLAB, which is intended for you to be able to work on the DSP system without relying on the lab facility
- To implement an embedded system that can work properly for RT DSP and achieve the desired performance, it is important and necessary to know the available resource and specifications of the system from the aspects of both hardware and software. Attention needs to be paid to
 - (i). the specifications on 32-bit AVR microcontrollers (in the hardware aspect) [M10], in particular, CPU, ADC, DAC, USART
 - (ii). the ASF (Atmel Software Framework) that are needed for programming [M11, M12]

Equipment, devices, hardware and software:

- a function generator, an oscilloscope, a multi-meter, a power supply
- the DIY circuit built during the first lab
- a computer installed with MATLAB, Atmel Studio 7, Flip and Putty; Use of your own laptop is highly recommended if you would like to work on the embedded systems at home as well
- A software framework in C (Lab-L2_DSP_framework.c) that includes ADC, DAC, external trigger and USART is available for implementing digital systems and performing RT DSP.
- C code (Lab-L2_USART_to_MATLAB_main.c) that includes data transfer from USART to MATLAB; and MATLAB code (Read_UC3.m) for receiving signals and data from the evaluation board via USART, which is used together with the C code.

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1. A digital system

A digital system (Fig. 1) can be described by an impulse response of $h(k)$ or a transfer function of $H(z) = Z\{h(k)\}$, and it has a digital input of $x(k)$ and a digital output of $y(k)$. Because most of the signals in nature (e.g., sound, pressure...) are analog, they need to be converted to digital signal through an analog to digital converter (ADC) in order for them to be suited for the digital system. The digital signals (e.g., music and songs that are stored in digital form in media storage, e.g., USB memories, are played back) need to be converted back to analog signals through a digital to analog converter (DAC).

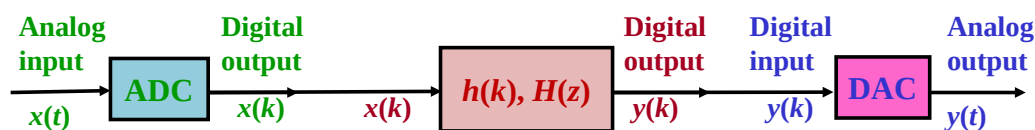


Fig. 1. A digital system that has an impulse response of $h(k)$ or a transfer function of $H(z) = Z\{h(k)\}$, a digital input $x(k)$, and a digital output of $y(k)$; and ADC and DAC.

2. Embedded system for DSP – AVR UC3-A3 Xplained evaluation kit

The embedded system used for real-time digital signal processing (RT DSP) in this course is the Atmel® UC3-A3 Xplained evaluation board (Fig. 2) that has an Atmel® AT32UC3A3256 microcontroller. The microcontroller has a 32-bit AVR32UC CPU, ADCs and DACs, that are sufficient for embedded DSP. The details on the microcontroller are available in [M10].

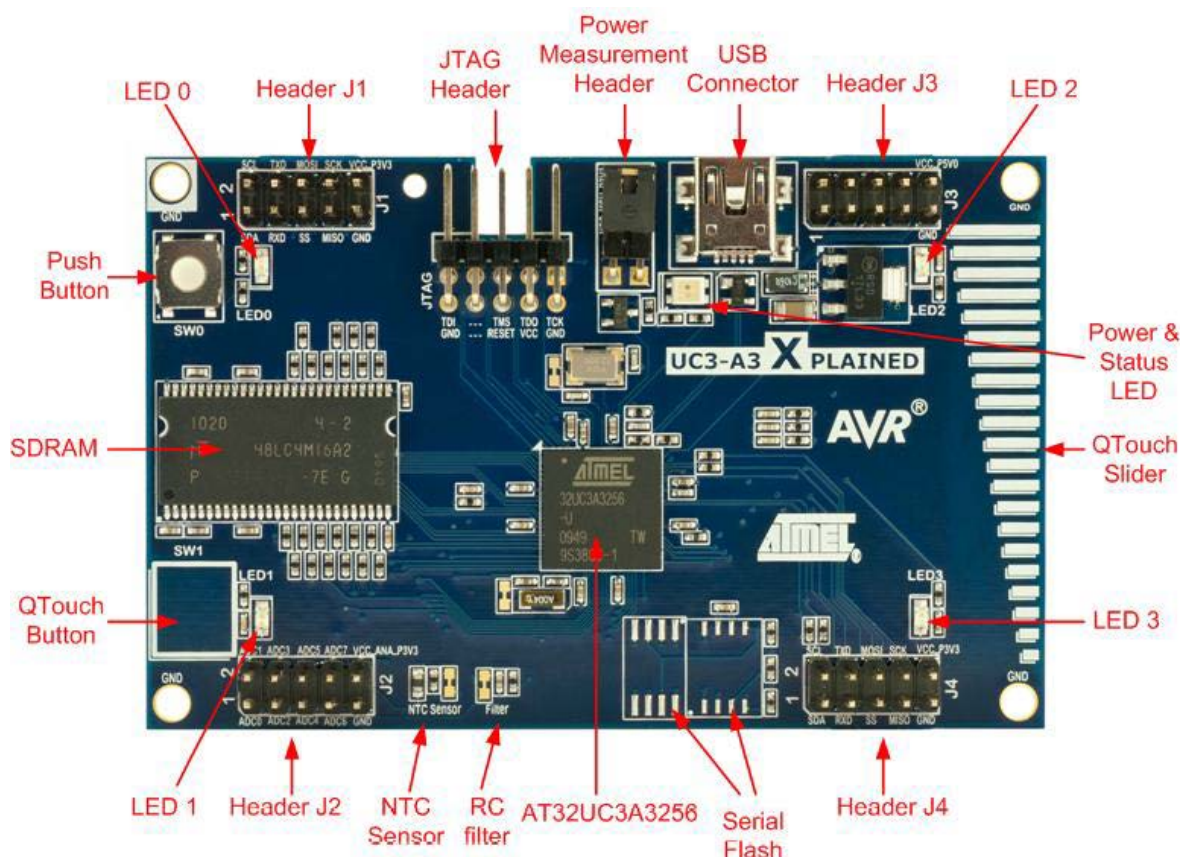


Fig. 2. Overview of the UC3-A3 Xplained evaluation board

To best use the embedded system for DSP and to be able to develop good-performance algorithms, it is necessary to know its available resource from the hardware and software aspects, and familiarize its specifications.

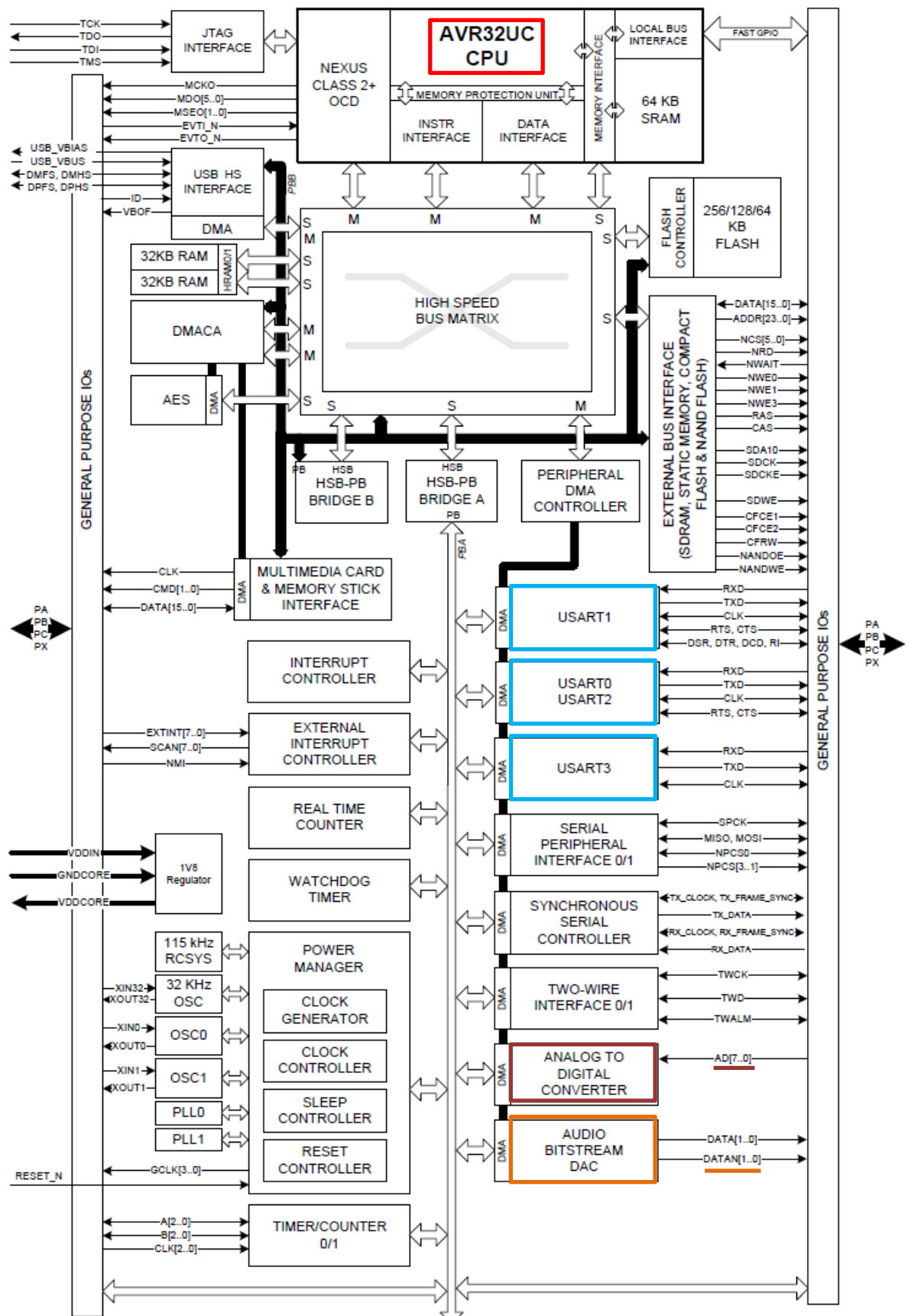


Fig. 3. Block diagram of AVR UC3 microcontroller [M10].

2.1 AVR UC3 microcontroller [M10]

The Atmel® AT32UC3A3/A4 is a complete System-On-Chip microcontroller based on the AVR32 UC RISC processor running at frequencies up to 66MHz. AVR32 UC is a high-performance 32-bit RISC microprocessor core, designed for cost-sensitive embedded applications, with particular emphasis on low power consumption, high code density and high performance. A DSP instruction set is included in the compact single-cycle RISC instruction set. The microcontroller has a 256 KB Flash, and a package of 144-ball TFBGA (Thin and Fine Ball Grid Array) with dimension 11x11 mm and pitch 0.8 mm. Its block diagram is shown in Fig. 3.

To the DSP purpose, the 8 channel ADCs (AD[7...0]), 2 channel Audio bitstream DACs (DATA[1...0]) and USARTs are of our main interest. Besides, the peripheral direct memory access controller (PDCA) is also of our interest, and used in the lab sessions to manage data transfer between ADC and memories and between memories and DAC. This is because PDCA enables data transfers between peripherals and memories without processor involvement. The PDCA drastically reduces processing overhead when transferring continuous and large data streams.

2.2 ADC

The AT32UC3A3 microcontroller embeds an Analog-to-Digital Converter (ADC) which is based on a Successive Approximation Register (SAR) 10-bit ADC. It integrates an 8-to-1 analog multiplexer, making possible the analog-to-digital conversions of 8 analog lines. The ADC is powered by an analog power supply, VDDANA, whose voltage is 3.3V nominal and may vary in the range of 3.0 – 3.6V. The conversions extend from 0V to VDDANA.

The ADC supports an 8-bit or 10-bit resolution mode, and conversion results are reported in a common register for all channels, as well as in a channel-dedicated register. Software trigger, external trigger on rising edge of the TRIGGER pin, or internal triggers from timer counter output(s) are configurable. The TRIGGER pin may be shared with other peripheral functions through the I/O Controller. The ADC also integrates a sleep mode and a conversion sequencer and connects with a PDCA channel. These features reduce both power consumption and processor intervention.

The 8 channel ADCs, ADC0 – ADC7, are available and accessible from header J2 on the UC3A3 Xplained board (Fig. 3). But they are functioning as ADCs only when they are configured and activated in the corresponding program. Some of the ADCs can be used to other purposes, for instance, ADC2 can be used as a DAC and any of the ADCs can be used as an external trigger, when they are configured for those purposes in the program.

Two particularly important ADC parameters are the *sampling rate* (or sampling frequency) which is configurable, and the *10-bit resolution* mode that is used in the lab sessions.

2.3 DAC

The AT32UC3A3 microcontroller has an audio bitstream DAC that converts a 16-bit sample value to a digital bitstream with an average value proportional to the sample value. Two channels are supported, making the audio bitstream DAC particularly suitable for stereo audio. Each channel has a pair of complementary digital outputs, DATAn and DATAnN, which can be connected to an external high input impedance amplifier. The DAC is also powered by VDDANA whose voltage is 3.3V nominal and may vary in the range of 3.0 – 3.6V. The conversions extend from 0V to VDDANA.

As configured in the provided software framework, the two audio bitstream DACs, DATA[1] and DATA[0], can be accessed from the pins, J2's ADC2 and J4's SCK on the evaluation kit.

Because the DACs have a 16-bit resolution and the ADCs have a 10-bit resolution, conversion of the ADC data to 16-bit data is needed.

2.4 USART

USART can be used for data communication between the MCU and PCs. Two USARTs on the UC3A3 Xplained evaluation board are accessible from J4 and J1. USARTs are powered by VDDIO, the I/O power supply whose voltage is 3.3V nominal and may vary in the range of 3.0 – 3.6V. **The USART in J4 is used in this lab session.**

3. Software framework for embedded DSP

A software framework for embedded DPS is provided. It includes ADC, DAC, USART, and external trigger. It is saved in a file with the name, Lab-L2_DSP_framework.c. A brief instruction is given in the beginning of the file. it is highly recommended to read it before you start to work on the project.

In the Lab session 1, it has been learnt how to use Atmel Studio 7 to write and build a project (program) and then use BatchISP to load the program to the MCU. In this lab session, you simply follow the procedure that you learnt in Lab session 1, and create a project, e.g., called ADC_DAC_USART, that implements the ADC, DAC, and USART.

To support ADC, DAC and USART, one needs to include the following services and drivers (Fig. 4):

- ABDAC – Audio Bitstream DAC (driver)
- ADC – Analog to Digital Converter (driver)
- DSPLib – Digital signal processing library (library)
- PDCA – Peripheral DMA Controller (driver)
- PM – Power Manager (driver)
- TC – Time/Counter (driver)
- USART – Universal Synchronous/Asynchronous Receiver/Transmitter (driver)
- System Clock Control (service)

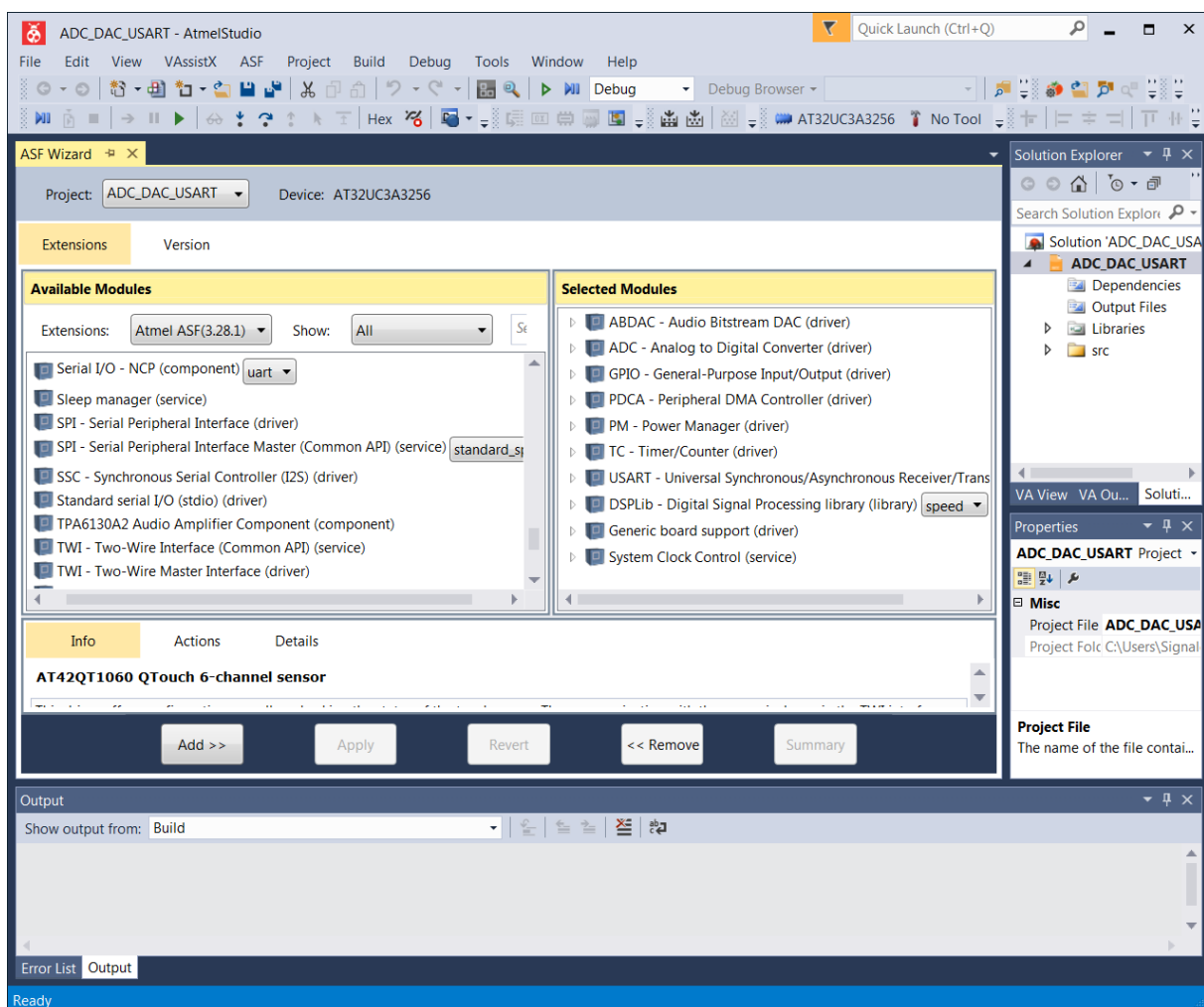


Fig. 4. The services and drivers (in the ASF) needed to include into the program/project.

4. Setups

The setup used in this lab session is shown in Fig. 5. The connections of the DIY circuit shown in the figure are the following (please refer to the file *Lab_1_DIY_Circuit.pdf* on studentportalen for the specification/naming of connectors)

- P5 → J3's Vcc and GND
- *P1 → J2's ADC1: ADC #0 (or J2's ADC5: ADC #1) with the RC LP filter
- or
- *P2 → J2's ADC1: ADC #0 (or J2's ADC5: ADC #1) without the RC LP filter
- P4 → J4's SCK: DAC #0 (or J2's ADC2: DAC #1)
- The USB-RS232 cable's TXD → J4's TXD
- The USB-RS232 cable's RXD → J4's RXD
- JK1 or P6 → analog input
- JK2 or P9 → analog output

*Note: P1 and P2 are connected to ADC #0 (or ADC #1), respectively, in order to see how aliasing presents and how the RC LP filter works.

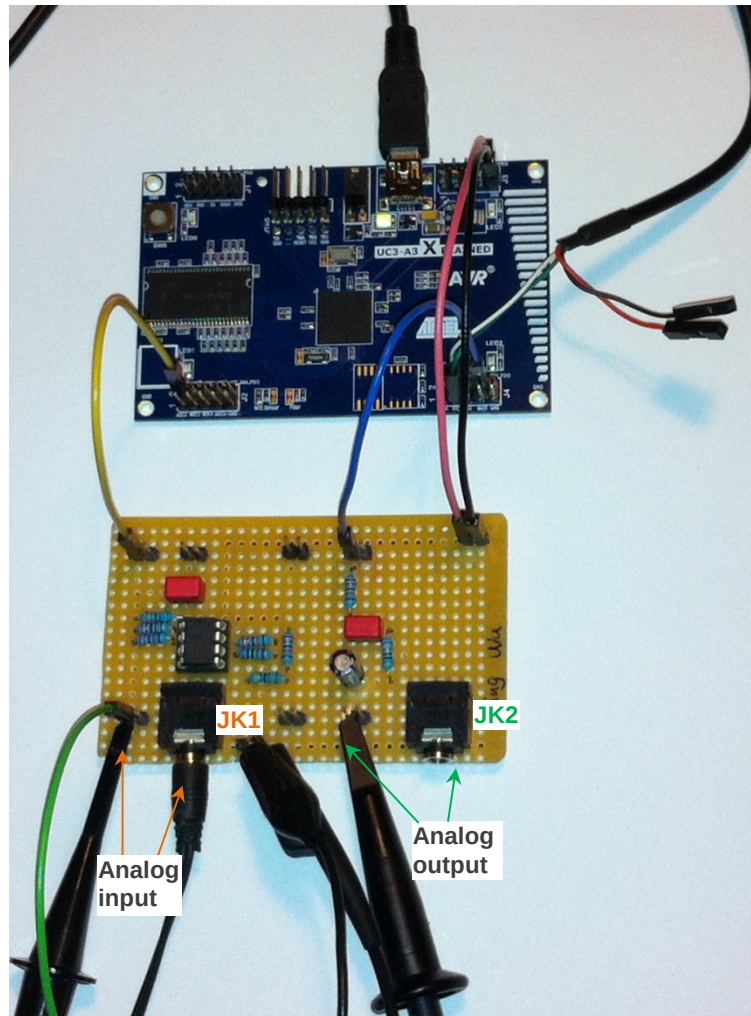


Fig. 5. Setup for lab experiments on embedded DSP

When the project ADC_DAC_USART is running on the evaluation board, the output can be obtained ONLY when the external trigger is activated (which can be made just by touching J2's ADC4 with your finger). Every time a change is made on the input (e.g., the frequency is changed), a touch on J2's ADC4 (a trigger) is needed.

5. Signal generation and USART data acquisition in MATLAB

MATLAB can be a good signal generator that can be used to generate a signal with an arbitrary waveform in the audio frequency range. For example, to generate a sine signal one can use the following codes

```
% Generate a 1 kHz sine waveform with a 10 seconds duration
Fs = 44.1e3;      % one of the standard sampling freq. for audio signals
Tm = 10;         % max duration in seconds
Nm = Fs*Tm;      % the max number of sampling points
t = (0:Nm)/Fs;   % t-axis
A0 = 1;          % Amplitude of the signal
f0 = 1e3;        % frequency of the signal in Hz
x_t = A0*sin(2*pi*f0*t);
nn = 1:200; plot(t(nn),x_t(nn))
sound(x_t, Fs); %sends x_t with FS out to the speaker. Note -1.0<=x_t<=1.0
```

To get the signal, use a cable with 3.5 mm plugs and plug in one end to the computer's earphone jack , and then get it on the other end. In this way, one can easily generate rectangular waveforms, triangular waveforms, and a sum of sine waveforms with different frequencies.

Note that the sound volume must be set to an appropriate level so the signal's amplitude is less than 1.6 volts or 3.2 volts peak-peak because of the limitation of the ADC's input. To make sure of this, check the signal level using an oscilloscope.

To receive the data from the UC3-A3 Xplained board through the COM port, you need to use MATLAB code, *Read_UC3.m*, on the PC and use the C code, *Lab-L2_USART_to_MATLAB_main.c*, on the Xplained board.

6. Tasks and exercises

The main task of this lab session is to study sampling theory and aliasing through lab experiments. In particular,

- (i). Observe how the output signal from the system changes (especially in frequency) when the input is a sine signal and its frequency is changed from 1 kHz – 50 kHz in two cases: (i) without and (ii) with an RC LP filter before the ADC. **Determine the sampling frequency of the DSP system.**
- (ii). Understand how the ADC and DAC work
- (iii). Think about how you implement an embedded DSP system when you have evaluation kit
- (iv). Generate signals using MATLAB, which is intended for you to be able to work on the DSP system without relying on the lab facility

Besides, the following exercises are recommended to work

- (1) Input a 1 kHz sine signal to the evaluation board, and then send it out to MATLAB via USART; After the MATLAB receives the whole data, plot it and determine its amplitude and frequency in MATLAB
- (2) use MATLAB to generate the following signal with a sampling frequency of 44.100 kHz
$$x(t) = \sin(2\pi f_1 t) + \sin(2\pi f_2 t) + \sin(2\pi f_3 t);$$
where $f_1 = 1$ kHz, $f_2 = 10$ kHz, $f_3 = 40$ kHz then use FFT in MATLAB to determine these frequencies and compare them with the original ones.

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